

KEVIN REN

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RESEARCH INTERESTS

My research is broadly focused on the intersection between AI and societal systems, developing interpretable and reliable methods to understand and improve the behavior of machine learning models in real-world settings. Currently, I apply these approaches to large language models (LLMs), characterizing and steering their generations to promote trustworthy deployment, with applications to academia and social media.

EDUCATION

Cornell University

Ph.D. student in Computer Science

- Advisor: Prof. Nikhil Garg

August 2025 -

New York, NY

Carnegie Mellon University

Bachelor of Science in Statistics and Machine Learning, Minor in Computer Science

- GPA: 4.0/4.0

- Extracurricular Involvement: Sustainable Earth (Treasurer), Asian Students Association (Secretary)

August 2021 - May 2025

Pittsburgh, PA

PEER-REVIEWED PUBLICATIONS

- **Predicting Language Models' Success at Zero-Shot Probabilistic Prediction.**
Ren, K, Cortes-Gomez, S, Patiño, CM, Joshi, A, Lyu, R, Tang, J, Turcan, A, Yamin, K, Wu, S, Wilder, B
Findings of Empirical Methods in Natural Language Processing (**EMNLP**), 2025.
- **Work Smarter Not Harder: Simple Imitation Learning with CS-PIBT Outperforms Large Scale Imitation Learning for MAPF.**
Veerapaneni, R, Jakobsson, A, **Ren, K**, Kim, S, Li, J, Likhachev, M
International Conference on Robotics and Automation (**ICRA**), 2025.
- **Decision-Focused Evaluation of Worst-Case Distribution Shift.**
Ren, K, Byun, Y, Wilder, B
Conference on Uncertainty in Artificial Intelligence (**UAI**), 2024.
- **Improving Learnt Local MAPF Policies with Heuristic Search.**
Veerapaneni, R, Wang, Q, **Ren, K**, Jakobsson, A, Li J, Likhachev, M
International Conference on Automated Planning and Scheduling (**ICAPS**), 2024.

RESEARCH EXPERIENCE

Feature Selection in Deep Learning, Yale University

Advised by Prof. Yuval Kluger, Prof. Ofir Lindenbaum

- Researched feature selection methods for hypernetworks, implementing sparse gating network in the Moethernet hypernetwork architecture.

June 2025 - August 2025

New Haven, CT

Resource Allocation and LLM Uncertainty, CMU Lab for AI and Social Impact

Advised by Prof. Bryan Wilder, Prof. Zhiwei Steven Wu

- Researched methods inspired by decision-focused learning for evaluating worst-case distribution shifts in resource allocation settings, given a predictive model, set of individuals, and downstream loss function.
- Investigated ability of Large Language Models (LLMs) to forecast their own performance on tabular data classification tasks, prior to observing ground truth labels.

May 2023 - May 2025

Pittsburgh, PA

Multi-Agent Pathfinding, CMU Search-Based Planning Laboratory

Advised by Prof. Maxim Likhachev

- Studied Graph Neural Networks (GNNs) for the Multi-Agent Pathfinding (MAPF) problem.
- Integrated machine learning methods such as GNNs and convolutional networks with heuristic MAPF solvers (La-CAM, PIBT), outperforming SOTA machine learning solutions.

March 2023 - May 2025

Pittsburgh, PA

Economics and Data Science, Metro21: Smart Cities Institute

Advised by Karen Lightman, Prof. Rick Stafford

August 2022 - May 2023

Pittsburgh, PA

- Analyzed regional economic health using municipality employment, financial, and education metrics
- Performed data engineering in Python and R to generate insights from regional poverty and transportation data for National Science Foundation proposal for regional economic stimulation funding

TEACHING EXPERIENCE

Cornell Tech

August 2025 - December 2025

Teaching Assistant, Applied Data Science - Decision-making beyond Prediction (ORIE 5355)

New York, New York

- Manage course infrastructure for a master's level class in applied data science
- Write code and descriptions for assignments (A/B testing, pricing algorithms, survey data analysis)
- Manage a team of graders to host office hours and grade assignments

Carnegie Mellon School of Computer Science

January 2023 - May 2025

Teaching Assistant, Introduction to Machine Learning (10-601, 701)

Pittsburgh, PA

- Led maintenance and feature development on course-hosted office hours queue web application.
- Hosted recitations and office hours, helping students with topics from linear regression to recommender systems.
- Collaborated with other TAs and course faculty to develop a new deep learning coding assignment, constructing RNNs from scratch in Pytorch.

INDUSTRY EXPERIENCE

Simplehuman, LLC

March 2026 - Present

Consultant

Remote

- Providing expertise on foundation models for upcoming product.

TikTok

May 2024 - November 2024

Machine Learning Engineer Intern

San Jose, CA

- Developed pipeline for fine-tuning, hyperparameter optimization, and data augmentation for LLMs for time series forecasting, achieving 20% higher accuracy.
- Pushed 2 new neural-network-based forecasters to production.
- Integrated new forecasting components into advertisement inventory system after presenting data analysis to 20+ person development group, reducing percent error by 40%.
- Researched business-focused metrics to improve forecasting evaluation.

Fifth Third Bank

May 2023 - August 2023

Software Engineer Intern, Secure Development Team

Cincinnati, OH

- Developed client-facing NodeJS, AWS-hosted microservice to onboard internal apps to encryption service.
- Configured and wrote Java (Spring Boot) tokenizer API to give clients example use cases of tokenization.

AWARDS

- NSF Graduate Research Fellowship Program (GRFP) Awardee, April 2026
- Phi Beta Kappa, May 2025
- CMU Senior Leadership Recognition, May 2025
- Dietrich College Dean's List, High Honors (Spring/Fall 2021-2025)
- Summer Undergraduate Research Fellowship, Summer 2022
- Awarded cash prize from CMU Statistics Department, 1st place in class project competition for Statistical Graphs and Visualization (link), December 2022

SKILLS / INTERESTS

Theoretical: Distribution Shift, Robustness, Deep Learning, Time Series Forecasting, Multi-Agent Pathfinding

Software: Python, C, C++, SQL, JavaScript (React, NodeJS), Java (Spring Boot, Gradle), Rust, R, Bash, Excel

Libraries: PyTorch, NumPy, Pandas, Matplotlib, PySpark, SciPy, SciKit-learn, Seaborn, SciPy

Technologies: AWS, Git, VSCode, Splunk, Jenkins, Docker, Terraform, HashiCorp Vault, Linux, PowerShell